BTS (C) – V – 10 – 051 – L

B. Tech Degree V Semester Examination November 2010

ME 503 MECHANICS OF MACHINERY

(2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART - A

(Answer ALL questions)

(8 x 5 = 40)

I. (a) With neat sketch explain the functioning of "Whitworth Quick Return Mechanism".

(b) State and explain "Arnold – Kennedy's Theorem of Three centres".

(c) Briefly explain the Forward and Inverse kinematics of Robots.

(d) What is Freudenstein's equation? How is it helpful in designing a four-link mechanism when three positions of the input $(\theta_1, \theta_2, \theta_3)$ and the output link (ϕ_1, ϕ_2, ϕ_3) are known?

(e) State and derive the "Law of gearing".

(f) Derive an expression for speed ratio of (i) simple gear train (ii) compound gear train.

(g) Deduce expressions for the velocity and acceleration of the follower when it moves with simple harmonic motion.

(h) Deduce an expression for the efficiency of an inclined plane when a body moves up a plane.

PART - B

(4 x 15 = 60)

II. (a) Sketch a Paucellier mechanism. Show that it can be used to trace a straight line. (7)

(b) The driving shaft of a Hooke's joint rotates at a uniform speed of 400 rpm. If the maximum variation in speed of the driven shaft is $\pm 5\%$ of the mean speed, determine the greatest permissible angle between the axes of the shaft. What are the maximum and the minimum speeds of the driven shaft? (8)

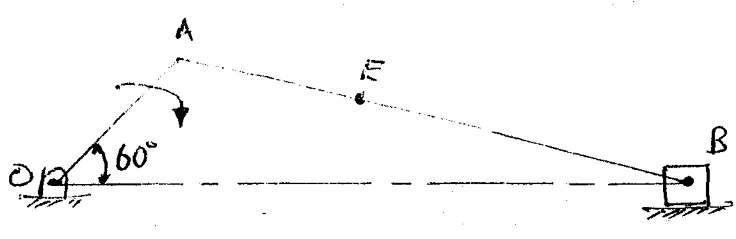
OR

III. For the configuration of a slider-crank mechanism shown in figure , draw velocity and acceleration diagram. Determine –

(i) the acceleration of the slider

(ii) the acceleration of point E

(iii) the angular acceleration of link AB.



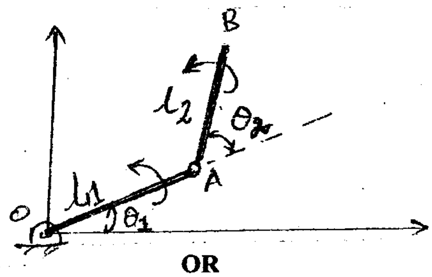
The crank OA rotates at 200 rpm clockwise.
OA = 500 mm; AB = 1500 mm; AE = 450 mm. (15)

IV. (a) Briefly explain about at least five possible robot coordinate frames. (5)

(P.T.O.)

(b) For a two-degree-of-freedom planar mechanism shown in figure.

$$\begin{bmatrix}\text{Differential}\\ \text{motion of B}\end{bmatrix} = \begin{bmatrix}\text{Jacobian}\end{bmatrix}\begin{bmatrix}\text{Differential}\\ \text{motion of joints}\end{bmatrix}$$, explain with derivation. (10)



OR

V. Design a four-link mechanism when the motions of the input and the output links are governed by a function $\boxed{y = x^2}$ and x varies from 0 to 2 with an interval of 1. Assume input link angle θ to vary from 50° to 150° and the output link angle ϕ from 80° to 160°. (15)

VI. (a) Make a comparison of cycloidal and involute tooth forms. (5)

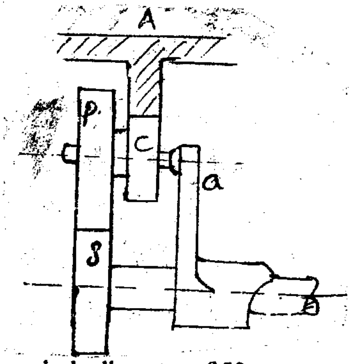
(b) A pair of spur wheels with 14 and 21 teeth are of involute profile and pressure angle 16°. Find maximum addenda on the pinion and gear wheel to avoid interference if module is 6 mm. Also find the maximum velocity of sliding on either side of the pitch point if pinion runs at 300 rpm. (10)

OR

VII. The number of teeth in the gear train shown in figure are as follows :

$T_S = 18,\quad T_P = 24,\quad T_C = 12,\quad T_A = 72$

P and C form a compound gear carried by the arm 'a' and the annular gear A is held stationary. Determine the speed of the output at arm a. Also find the holding torque required on A if 5 KW is delivered to 'S' at 800 rpm with an efficiency of 94%. In case the annulus A rotates at 100 rpm in the same direction as S, what will be the new speed of arm 'a'? (15)



VIII. A tangent cam with a base circle diameter of 50 mm operates a roller follower 20 mm in diameter. The line of stroke of the roller follower passes through the axis of the cam. The angle between the tangential forces of the cam is 60°, speed of the cam shaft 200 rpm and the lift of the follower 15 mm. Calculate –

(i) the main dimensions of the cam

(ii) the acceleration of the follower at

 (a) the beginning of lift

 (b) where the roller just touches the nose

 (c) the apex of the circular nose. (15)

OR

IX. (a) A body is to be moved up an inclined plane by applying a force parallel to the plane surface. It is found that a force of 3 kN is required to just move it up the plane when the angle of inclination is 10°, whereas the force needed increases to 4 kN when the angle of inclination is increased to 15°. Determine the weight of the body and the coefficient of friction. (7)

(b) In a screw jack, the diameter of the threaded screw is 40 mm and the pitch 8 mm. The load is 20 kN and it does not rotate with the screw but is carried on a swivel head having a bearing diameter of 70 mm. The coefficient of friction between the swivel head and the spindle is 0.08 and between the screw and nut 0.1. Determine the total torque required to raise the load and the efficiency. (8)

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